



The Evolution of Execution

ARM 1 Simulation & Pipelined ARM Implementation

Course: CIE 439 Computer
Architecture & Assembly

Project Objectives & Scope

Project 1: ARM 1 Simulation

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✓ **Objective:** Simulate the ARM 1 architecture.
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✓ **Method:** Hardware simulator (Logisim, SystemVerilog).
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✓ **Output:** Multi-cycle step execution.

Project 2: Pipelined ARM Processor

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✓ **Objective:** Implement a pipelined ARM processor.
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✓ **Method:** SystemVerilog.
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✓ **Output:** Supported instructions execution and handling of pipeline hazards.

Core ARM Architectural Background



RISC Architecture

ARM operates on a Reduced Instruction Set Computer architecture, prioritizing simple, fast instructions.



Strict Memory Access

Uses a pure Load/Store design. Memory is accessed exclusively via the `LDR` and `STR` instructions.



Data-Processing

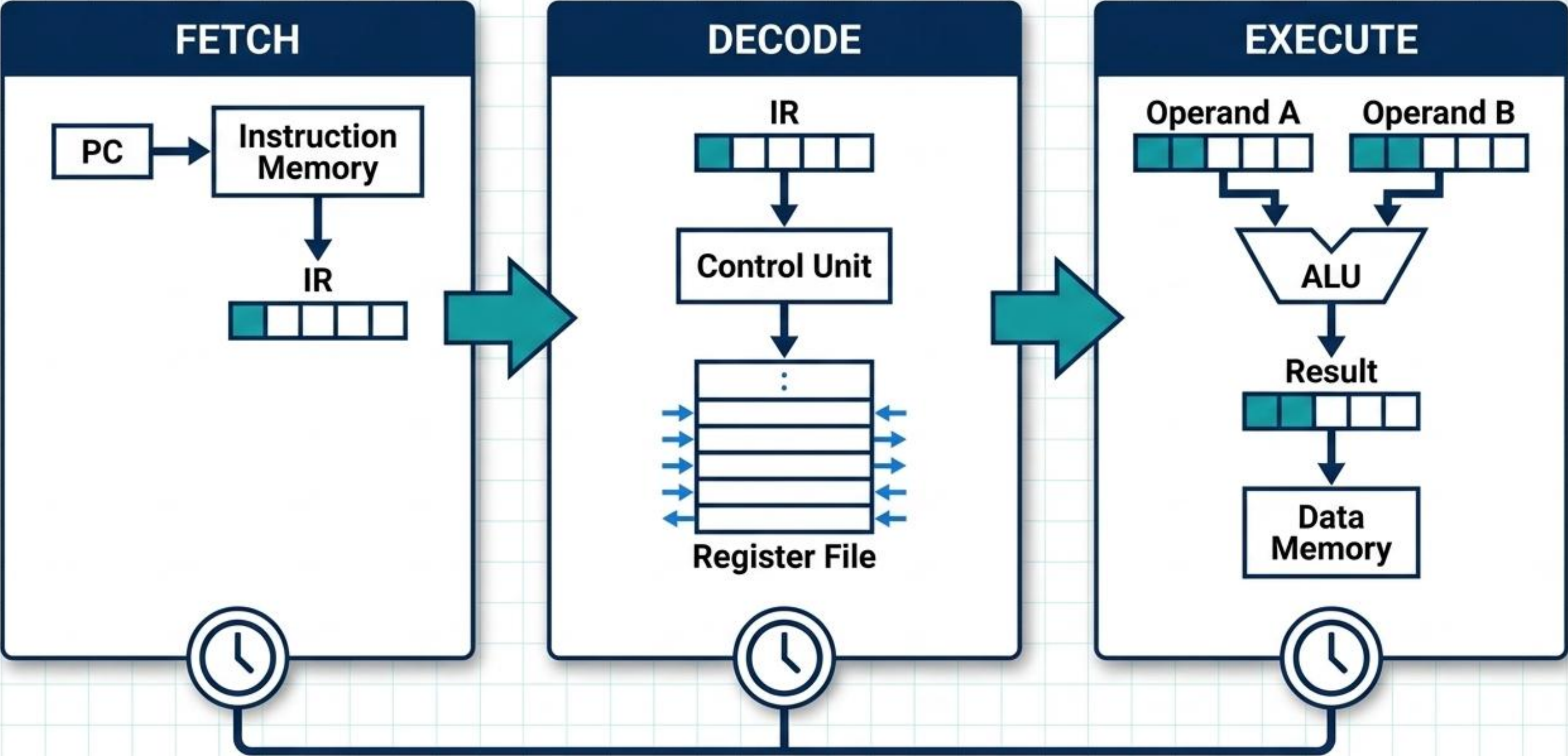
Arithmetic and logic operations run purely via dedicated data-processing instructions without direct memory mutation.

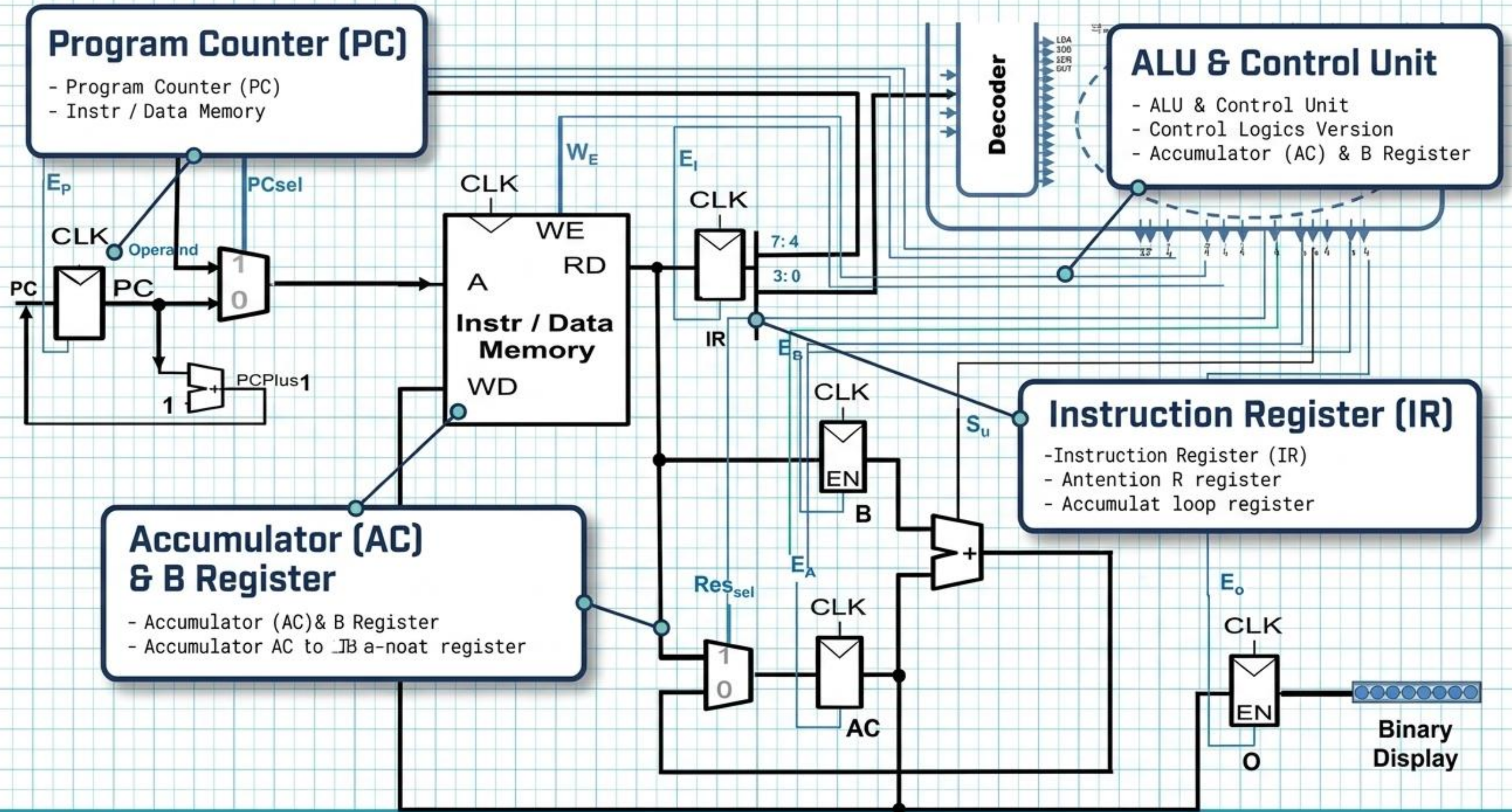


Hardware Division

The system is cleanly split into a Datapath (which performs operations) and a Control Unit (which orchestrates execution).

Project 1: Multi-Cycle Execution Flow





SystemVerilog Implementation Architecture

Implementation
Logic

Toolchain:
SystemVerilog
+ ModelSim

